APPEL A LETTRES D'INTENTION BIODIVOC CONSORTIUMS D'EQUIPES

FORMULAIRE DE CANDIDATURE

1. IDENTIFICATION DU PROJET

Titre du projet	Soil **B**iodiversity and **E**cosystem mu**L**tifunctionality in **O**ccitanie under **W**ater constraints
Acronyme du projet	**BELOW**
Mots-clés (5)	Ecosystem functioning, Extreme climatic events, Greenhouse gas emissions, Multitrophic diversity, Soil organisms

2. RESPONSABLES ET UNITES DE RECHERCHE

	Responsable scientifique (nom, organisme, équipe, courriel)	**Unité / ville**
Unité 1*	Grégoire Freschet, CNRS, Linking, gregoire.freschet@sete.cnrs.fr	SETE / Moulis
Unité 2*	Stephan Hättenschwiler, CNRS, Bioflux, stephan.hattenschwiler@cefe.cnrs.fr	CEFE / Montpellier
Unité 3	Antoine Lecerf, Université de Toulouse, EcoBIZ, antoine.lecerf@univ-tlse3.fr	Laboratoire d'Ecologie Fonctionnelle et Environnement (LEFE) / Toulouse
Unité 4	Alexandru Milcu, CNRS, Ecotron, alex.milcu@ecotron.cnrs.fr	ECOTRON / Montpellier
Unité 5	Alain Brauman, IRD, alain.brauman@ird.fr	ECO&SOLS / Montpellier

*** Les deux unités 1 et 2 vont co-porter le projet comme porteurs principaux ensemble**

3. PARTENAIRES EXTERIEURS (le cas échéant)

	Nom / localisation	**Type de structure**	**Responsable (nom, téléphone, courriel)**
Partenaire 1	CESBIO / Toulouse	Unité de recherche	Vincent Bustillo, 05 61 55 85 01, vincent.bustillo@iut-tlse3.fr
Partenaire 2	G-EAU / Montpellier	Unité de recherche	Claire Serra-Wittling, 04 67 04 63 12, claire.serra-wittling@inrae.fr
Partenaire 3	EMMAH / Avignon	Unité de recherche	Yvan Capowiez, 04 32 72 24 38, yvan.capowiez@inrae.fr
Partenaire 4	ADEME – Service Agriculture et Forêt	Agence	Thomas Eglin, 02 41 91 40 55, eglin@ademe.fr

4. DESCRIPTION DU PROJET

4.1 *Résumé du projet*

Soils are a critical ecosystem component for climate change mitigation and ecosystem functioning and need to be better incorporated in biodiversity conservation plans. Here we address the questions of how the diversity of key groups of soil organisms drive multiple ecosystem functions (greenhouse gas emissions, soil C storage, nutrient cycling) and how it is affected by changes in water availability (extreme drought, flooding). Controlled experimental manipulation, field measurements and modelling are combined in a multiscale, mechanism-driven approach covering a variety of ecosystem types: agricultural fields, grassland, peat bog, wetland, and Mediterranean forest. The sharing of unique infrastructure (instrumented field sites and ecosystem-scale experiments) and state-of-the-art methodology by this interdisciplinary consortium can foster a new and durable research network on soil functional biodiversity within Occitanie and provide expertise for future decisions in land management.

4.2 *Contexte scientifique et objectifs*

- Etat de l'art ;

Soils play a key role in terrestrial ecosystems by driving those processes that underpin the provision of key ecosystem goods and services including soil fertility, control of the water cycle, plant productivity and climate regulation (1-3). There is strong concern about the soils' capacity to serve as a carbon (C) stock, water filter, and support of food and fiber production with increasing pressure by ongoing climate change, land use change and the strong modifications of biodiversity (4-6). An improved and integrative soil management across different ecosystems types ranging from agricultural ecosystems, to natural ecosystems such as grasslands, forests and wetlands accounting for predicted climate change is urgently needed and should be a major issue in environmental policy (7).

Soils harbour an astounding diversity of organisms across essentially all branches of the tree of life (3), which is organized in complex, mostly trophic, interaction networks (8, 9). The enormous diversity and associated functional characteristics of soil organisms such as microorganisms (10, 11), soil fauna (12-14) or plant roots (15) and how they contribute to soil processes, makes a mechanistic understanding and predictive framework very challenging even within single trophic levels (16, 17). The characterization of the functional structure of communities, i.e. the composition and diversity of the functional traits of organisms (18), is a powerful approach that has been extensively developed in the past decades in order to link organisms to the role they play in ecosystems. For a given trophic level, higher trait diversity is usually considered to increase the rates of ecosystem processes, because functionally distinct organisms may exploit distinct resources (16). However, soil functions depend on the functional diversity of several interacting trophic levels, advocating for a multitrophic functional characterization of soil biota (19-22). Linking multitrophic soil functional diversity to multiple functions in a diversity of agro- and natural ecosystems could therefore provide a firm basis for decision making in ecosystem management policies (23).

In southern France, substantial reductions of precipitation and increasing intensities of summer drought are expected for the second half of the 21st century (24-26). At the same time, the intensity of single rainfall events causing local flooding is predicted to increase (27). Altered climatic conditions impact soil processes in a complex way, with both direct (e.g. water availability), and indirect effects, by changing the composition of biological communities. A mechanistic approach is needed to disentangle these effects and anticipating how the relation between soil biodiversity and functions will be modified. Microbial community responses to altered rainfall distribution may vary (28). Generally, while soil microbial diversity may decrease in response to increasing aridity (29), the fungal community seems more resistant to drought than bacteria (30, but see 31), and drought can alter functional gene abundances (32). Such changes in microbial communities can affect ecosystem process rates (33, 34). Soil fauna may respond differently to drought compared to microorganisms, being particularly sensitive to the frequency of rainfall events rather than the absolute amount of rainfall (35, 36). However, drought tolerance and resistance vary among species of soil fauna and may further depend on the functional diversity of communities. Plants also show strong adaptations of their root traits (37), with numerous potential consequences for soil and ecosystem functioning (17). For example, changes in root exudate quantity and quality after drought can lead to increased decomposition of soil organic C (priming) and contribute to the sharp peak in CO_2 production when soil is rewetted (38). Overall, it is largely unknown how the response of soil microorganisms, soil fauna and plants are connected, and how they modulate the functioning of soils and entire ecosystems with ongoing climate change and loss of biodiversity. Because soil biota control much of the greenhouse gas emissions, potential feedback or feedforward effects could be substantial with large consequences on biosphere – atmosphere interactions.

- Questions posées et objectifs ;

The overarching questions of the proposed research are: **How are ecosystem processes regulated by multiple levels of soil biodiversity across contrasting ecosystem types of southwestern France and how are these linkages influenced by climate change ?**
The main objective is an improved general understanding of the role of soil biodiversity in driving ecosystem multifunctionality including processes relevant for nutrient cycling, greenhouse gas emissions, soil carbon storage, and primary productivity across different ecosystems. A major aim is the quantification of the relative importance of soil biodiversity compared to environmental drivers in order to better predict climate change impacts at the landscape level. Because anticipated changes in rainfall patterns and water availability are a major threat in the studied region, there will be a strong focus on these components of climate change and how soil biodiversity may potentially mitigate these effects. A second important general objective is to translate the insights of how soil biodiversity regulates ecosystem processes to applicable tools for land managers. To do this we intend to propose a selection of biological metrics that are good proxies for soil functions and that are also sensitive to climate change, which may help to monitor how climate change will impact ecosystem functions and services. These indicators could be used in land management plans for explicit recommendations for ecosystem management and end-users.

- Méthodologie ;

A combination of observational, experimental, and modelling approaches covering different spatial scales constitutes the methodological backbone of the proposal. Five, already instrumented field sites (two of them including an experimental manipulation of water availability), covering different types of ecosystems (two agroecosystems (one near Toulouse and one near Montpellier) and three sites with natural ecosystems: Pyrenean peatbog, alluvial wetland and Mediterranean forest) will be used to characterize landscape-scale variation in soil biodiversity focussing on four major groups of organisms (microorganisms, nematodes, earthworms and (plant)roots), known for their particular strong impact on ecosystem processes. We will use molecular and functional approaches, and its covariation with environmental variables and a wide range of ecosystem functions such as nutrient and water cycling, soil carbon storage, decomposition, greenhouse gas GHG emissions (CO_2, N_2O, CH_4), and primary productivity. In addition, two mesocosm experiments (Mediterranean grassland and forest, respectively) manipulating soil biodiversity and precipitation patterns permit to explore the underlying mechanisms of how soil biodiversity, climate and ecosystem functioning are interrelated. *In silico* modelling of the effects of predicted climate change and its consequences at the landscape level will be conducted based on field and experimental data.

- Résultats attendus et livrables.

The expected data of unprecedented breadth and detail will allow to evaluate the role of multitrophic soil biodiversity for ecosystem multifunctionality (39), and in particular on biogeochemical cycling and GHG emissions across key ecosystems in the Occitanie. Moreover, the experimental and modelling assessment of the consequences of climate change (precipitation regimes, extreme events) in interaction with soil biodiversity will permit a better understanding of the future threats on key ecosystem properties and services and potential counteractions and mitigation strategies. We expect explicit deliverables for ecosystem management with regard to soil biodiversity conservation, soil carbon storage, and mitigation potential for GHG emissions in the face of climate change.

4.3 *Ambition du projet*

- Dimension novatrice et percées potentielles ;

The combined analysis of multitrophic soil biodiversity (especially the formal combination of absorptive root traits and key groups of soil organisms interacting with plants) and various functional approaches across different ecosystem types is highly novel, allowing a more detailed understanding of multiple dimensions of soil biodiversity and its functions. The parallel assessment of a wide range of ecosystem processes, in particular with a strong focus on carbon storage, GHG emissions, and biogeochemical modelling across spatial scales will further allow to link multiple levels of soil biodiversity and ecosystem functions and services more explicitly at the landscape level. A major strength of the project is the explicit test of climate change related extreme events (summer drought, flooding) and/or irrigation practice on ecosystem functioning in interaction with biodiversity, from which we expect specific outputs for ecosystem management that were inexistent so far.

- Dimension structurante pour la communauté scientifique de l'Occitanie ;

Presently, the research on soil biodiversity, climate change, and biogeochemical cycling is not as well structured regionally as it could be. The project would permit a collaboration across disciplines, approaches, scales of investigation, and study objects that so far remained largely separated (e.g. agricultural vs. natural ecosystems, plant vs. animal, and microbial communities, observational vs. experimental, and modelling approaches, understanding of mechanisms vs. patterns). Notably, some of the research teams involved in the project had no or minor collaboration in the past, and it would be the first time they would share their infrastructures to achieve a common goal. Clearly, the project would initiate a longer-term collaborative effort with a strong synergistic and lasting impact on the research community in the Occitanie, allowing a strong national and international positioning.

- Interdisciplinarité ;

  The soil, in essence, is a highly complex medium, hosting contrasting groups of organisms, each type with its own independent field of research and expertise. Here, we propose to bridge expertise from four major groups of organisms, microorganisms, nematodes, earthworms and plant roots, and the expertise of plant and soil ecologists, soil scientists, biogeochemists, ecosystem scientists and modellers in a strong interdisciplinary effort.

- Intérêt pour la gestion de la biodiversité (le cas échéant).

  The recognition of the importance of soil biodiversity lags strongly behind in management and conservation strategies compared to vertebrate, pollinator, or plant diversity. This represents a strong bias in ecological understanding and also in public and stakeholder perception. We believe that our research will have an important impact on the understanding of soil biodiversity as driver of ecosystem multifunctionality and will help to identify relevant indicators of ecosystem functioning and increase the awareness of the role of soil biodiversity. At two of the sites included in our project there are practical student (in agronomy and ecology) classes that contribute to such training and that will benefit from the output of our project in acquiring state-of-the art knowledge and skills in soil biodiversity research. Moreover, two of our field sites (peatbog and alluvial wetland) are part of nature conservation areas (PNR, RNR). The project results will be made accessible to managers and a wider public, which collectively should change the perception of soil biodiversity and its importance for land conservation purposes.

4.4 *Organisation du projet et moyens*

- Méthodologie, utilisation de plates-formes expérimentales et sites d'observation

  Field sites:
  Three observational sites are selected in the western Occitanie: a 600-ha nature conservation area within an alluvial wetland along the Garonne River ("Parc du confluent"), a highland peatbog in the Ariège department (Bernadouze) which has been studied for GHG (CO_2, N_2O and CH_4) emissions since 2016; a cropfield in the agricultural landscape of the Gers department. The latter two sites support long-term observation and ecological research since 2011 and 1982, respectively, and have been instrumented since then as part of several national research networks (OHM Pyrénées Haut Vicdessos-Labex DRIIHM & SNO Tourbière CNRS, OZCAR network, ICOS and ZA PYGAR). High resolution data are available for climatic and several soil factors and fine-scale spatial variations in water availability have been mapped, making these sites particularly precious for the questions asked here. Two more sites in the eastern Occitanie include an evergreen oak Mediterranean forest at Puéchabon, which is instrumented since 2001 as part of several national and international research infrastructures (ICOS ERIC, AnaEE RI, OSU OREME) where water and carbon budgets are continuously monitored along with meteorological and soil variables, and where a long-term rainfall manipulation experiment allows to simulate drier conditions since 2003. The Mediterranean agroecosystem (Lavalette, near Montpellier), which is also part of the OSU OREME, is an irrigated crop field, manipulating soil management practices (agro-ecological with soil conservation techniques and conventional tillage) combined with three water supply methods (rainfall, sprinkler irrigation and subsurface drip irrigation at 40 cm depth). Plots have been instrumented to monitor climatic factors and to study the influence of management practices on soil hydrodynamic properties and to optimize irrigation.

Controlled experiments:

At the European Ecotron we will use an ongoing experiment testing the interactive effects of functional diversity of soil fauna and increased summer drought on the functioning of a Mediterranean forest ecosystem with a total of 16 large mesocosms (surface area of $1m^2$ and a soil depth of 1m). With continuous measurements of gas fluxes and the water balance, it will be possible to quantify the treatment effects at the scale of model ecosystems.
At the SETE greenhouse platform, we will make use of an ongoing experiment composed of 410 large containers (0.2 m diameter x 0.6 m depth) with grassland ecosystems that manipulate soil microorganism diversity (high vs low diversity of mycorrhizal fungi and rhizobia), herbaceous plant diversity (1, 2 and 6 species) and soil water availability (drought vs control) conditions. Measurements of ecosystem functioning (plant biomass production, soil C content, N leaching, water infiltration rate) will be complemented with gas flux measurements, water level monitoring and functional diversity of soil microorganisms.

Combining field sites along gradients of water availability, *in situ* manipulation of water availability, and ecosystem-scale experiments under controlled conditions with drought treatments using a multifunctionality approach of ecosystem functioning and a detailed assessment of multitrophic soil biodiversity is unique. This is a powerful approach in the assessment of underlying mechanisms of the interactive effects of soil biodiversity and changing water availability on the functioning across a wide range of ecosystems in Occitanie.

Measurements and data analysis:

We will use a common methodology to acquire three types of data: i) genetic and functional diversity of soil organisms, ii) data on spatial and temporal variation in climatic variables, with a particular focus on water-related parameters), and iii) data on soil parameters (soil structure, carbon dynamics and nutrient availability) and ecosystem functions (nutrient cycling, carbon storage, greenhouse gas (GHG) emissions, and primary productivity). The diversity of microorganism will be assessed using a meta-barcoding approach with prediction of functional profiles (40, 41), and functional gene determination (42). Functional diversity of nematodes will be determined using the detailed functional trait database by J. Trap (NEMATRAIT), who participates in the consortium. Earthworm functional diversity will be assessed using morphological keys and the Betsi database (https://portail.betsi.cnrs.fr/), respectively. Root diversity will be characterized by functional traits of the most active absorptive roots (43). Collectively, these four groups of organisms are major determinants of the ecosystem functions evaluated during our project with well-identified trophic interactions.

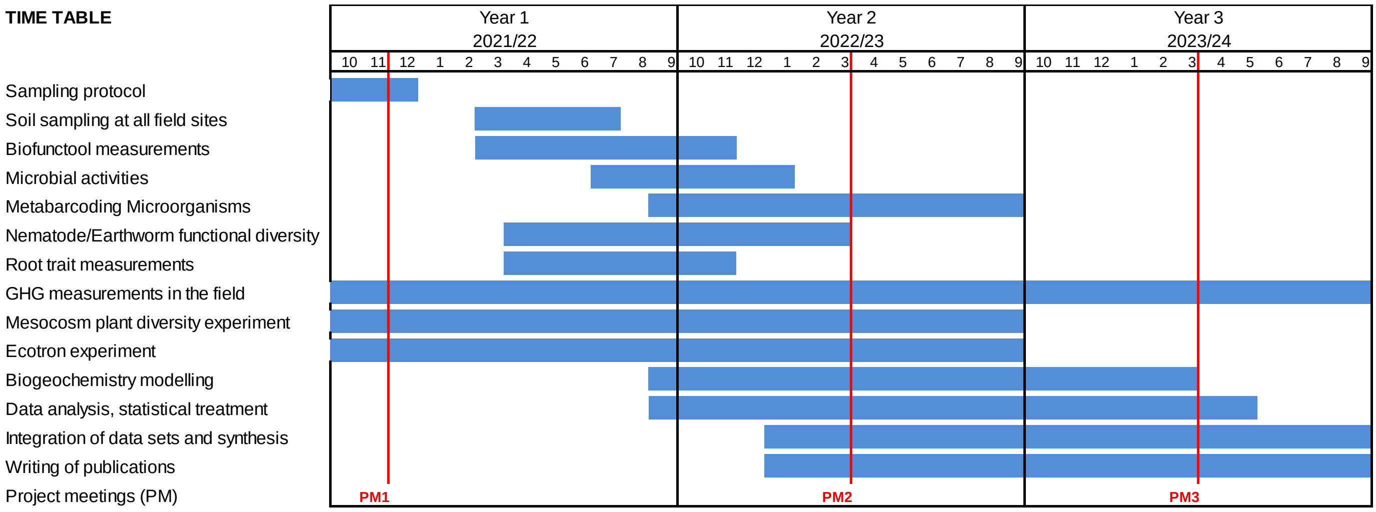
Greenhouse gas emissions will be measured at all five field sites across gradients of water availability for a better understanding of how fluctuating water availability affects these fluxes and how they depend on soil biodiversity. CO_2, CH_4 and/or N_2O emissions will be assessed *in situ* using closed-chamber systems connected to portable infra-red analyzers and *ex situ* under controlled conditions to assess the potential of microbial activities (44).

The microbial catabolic diversity will be assessed using the MicroResp® method (45). Soil functioning will be assessed in the field keeping the physical integrity of the soil. Three general functions (C dynamics, nutrient cycling, and soil structure, 46) will be targeted with the nine indicators used in the Biofunctool kit developed by ECO&SOLS (47).

Data collected with a common methodology from all sites will be used to test the relative importance of soil characteristics and soil biodiversity metrics on the multiple response variables related to ecosystem functioning. Generalized linear mixed effect models (GLMM) and additive mixed effect models (GAMM) will be used to account for site random effects and spatial and temporal autocorrelation. Structural equation modelling (SEM) will be used to test the strength of the causal relationships. Similar analyses will be performed on a global metric of ecosystem multifunctionality computed according to the threshold method (39, 48).

Furthermore, we will adopt a mechanistic modelling approach to better infer the role of multitrophic diversity on processes underpinning GHG emissions from soils. Biogeochemical models are able to produce estimates for emissions of trace gas relevant to the carbon and nitrogen cycles (e.g. N_2O, NH_3, CH_4, CO_2). We will use the most comprehensive model to date (i.e. the DeNitrification-DeComposition, DNDC) to investigate biotic and abiotic drivers across different ecosystem types in Occitanie, and to simulate different climate change scenarios.

- Calendrier prévisionnel ;



- Description et complémentarité des partenaires ;

The consortium is highly complementary with expertise in functional diversity of soil organisms (ECO&SOLS, CEFE), plant diversity (SETE, CEFE), root ecology (SETE), soil microbial ecology (ECO&SOLS, CEFE) biogeochemistry (Laboratoire d'Ecologie Fonctionnelle et Environnement, LEFE), greenhouse gas emissions (LEFE, ECOTRON), experimental facilities with running cutting-edge experiments (SETE, ECOTRON, CEFE), biogeochemical modelling (LEFE) and a general strong background in different facets of global change ecology and functional ecology (CEFE, ECO&SOLS, LEFE, SETE, ECOTRON) combining an exceptional breadth of competence and experimental facilities to address the questions exposed in this letter of intention. In addition, due to their leading role the different partners are strongly connected internationally.

Apart of ongoing collaborations between CEFE and the ECOTRON none of the other partners were or are collaborating formally with each other.

The external partner CESBIO brings expertise in the field of greenhouse gas measurements. The colleagues from G-EAU support the project by their experimental site and with their expertise in soil hydrology. The colleagues from EMMAH are experts on earthworm ecology and soil microbial activity under climate change. Finally, the extra-academic partner at ADEME is interested in the results and how our case studies in Occitanie can contribute to growing national efforts to promote sustainable soil management.

- Budget global, répartition annuelle et entre unités ;

Year	SETE	CEFE	LEFE	ECOTRON	ECO&SOLS	**TOTAL**
1st	16'450	23'250	46'500	16'000	23'300	**125'500**
2nd	11'550	81'250	31'200	16'000	20'600	**160'600**
3rd	1'000	37'000	8'600	-	1'000	**47'600**
Total	29'000	141'500	86'300	32'000	44'900	**333'700**

A co-lead of the project, yearly workshops and a shared Postdoc across all labs (and based in Montpellier) will be instrumental in facilitating exchanges across labs. Several Technicians will also be hired at the interface between labs.

None of the five responsible persons are presently funded through a large national or international project.

	Nom et courriel DU, date, signature
Unité 1	Michel Loreau, michel.loreau@sete.cnrs.fr
Unité 2	Marie-Laure Navas, marie-laure.navas@cefe.cnrs.fr
Unité 3	Régis Céréghino, Regis.cereghino@univ-tlse3.fr
Unité 4	Alexandru Milcu, alex.milcu@ecotron.cnrs.fr
Unité 5	Laurent Cournac, 12/05/2021 laurent.cournac@ird.fr

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